

MILANO MALPENSA, Italy

WMO: 160660

Lat: 45.631N

Lon: 8.728E

Elev: 234

StdP: 98.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.9	(c) -5.9	(d) -14.8	(e) 1.1	(f) 2.9	(g) -12.1	(h) 1.4	(i) -0.3	(j) 8.9	(k) 10.1	(l) 7.1	(m) 8.3	(n) 0.4	(o) 0	(p) 0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.1	21.5	30.9	21.0	29.7	20.5	23.0	29.8	22.2	28.6	21.5	27.6	2.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.1	25.8	20.1	15.2	25.4	19.2	14.4	24.5	69.2	29.2	66.1	28.5	63.3	27.7	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.5	5.1	4.2	-10.9	34.0	2.3	1.4	-12.6	35.0	-13.9	35.8	-15.2	36.5	-16.9	37.5
			-11.5	24.2	2.2	1.4	-13.1	25.2	-14.4	26.0	-15.6	26.8	-17.2	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	2.3	3.7	8.1	12.0	16.6	20.9	23.1	22.6	18.1	13.1	7.4	2.9		
	DBStd	7.95	2.86	3.07	3.19	3.03	2.89	3.08	2.43	2.69	2.99	3.25	3.28	3.14		
	HDD10.0	820	239	176	73	14	1	0	0	0	0	10	88	219		
	HDD18.3	2504	498	408	317	190	70	11	1	2	39	162	328	477		
	CDD10.0	1777	0	0	16	74	204	326	406	391	243	107	10	0		
	CDD18.3	420	0	0	0	1	15	87	148	136	33	1	0	0		
	CDH23.3	3978	0	0	3	15	172	848	1453	1223	251	11	0	0		
	CDH26.7	1262	0	0	0	2	27	257	509	423	43	1	0	0		
(14) Wind	WSAvg	1.5	1.2	1.3	1.7	1.8	1.8	1.7	1.7	1.6	1.4	1.2	1.3	1.1		
(15) Precipitation	PrecAvg	1087	72	82	99	82	122	94	65	98	89	119	122	55		
	PrecMax	1689	206	268	279	227	332	194	160	240	265	337	225	129		
	PrecMin	717	2	0	0	1	22	29	14	7	1	0	8	5		
	PrecStd	297	52	67	74	56	84	51	39	58	73	102	68	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	17.8	23.0	25.0	29.1	33.1	33.9	34.1	29.7	24.5	18.0	14.0		
		MCWB	7.8	9.4	12.1	15.5	18.4	20.9	22.3	22.4	19.9	17.4	12.1	7.7		
	2%	DB	11.7	14.1	19.9	22.9	27.1	31.1	32.0	32.2	28.0	22.2	15.9	11.5		
		MCWB	6.4	7.6	11.2	14.3	17.7	20.9	21.4	21.6	19.2	15.7	11.1	7.2		
	5%	DB	9.8	12.2	17.9	21.2	25.1	29.8	30.8	30.7	26.2	20.8	14.1	10.0		
		MCWB	5.4	6.6	10.3	13.2	16.7	20.5	21.0	21.1	17.9	15.3	10.5	6.7		
	10%	DB	7.9	10.2	15.9	19.4	23.4	28.0	29.6	29.1	24.6	19.0	12.9	8.5		
		MCWB	4.6	5.6	9.5	12.3	15.8	19.5	20.5	20.2	17.2	14.3	10.3	6.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	9.9	12.9	16.2	20.0	23.7	24.1	23.9	21.2	18.5	13.6	10.1		
		MCDB	13.2	15.7	19.8	23.4	26.1	30.6	30.4	30.5	26.8	21.2	15.9	10.7		
	2%	WB	7.3	8.5	11.9	14.8	18.4	22.3	23.1	22.9	20.2	17.3	12.7	8.5		
		MCDB	9.8	12.4	17.8	21.1	25.1	29.0	29.6	29.8	25.7	20.2	14.5	10.0		
	5%	WB	6.0	7.4	11.0	13.9	17.4	21.2	22.3	22.1	19.4	16.4	11.7	7.4		
		MCDB	8.2	10.5	16.1	19.7	23.7	27.8	28.5	28.5	24.5	19.1	13.4	9.0		
	10%	WB	5.0	6.4	10.1	12.9	16.6	20.2	21.5	21.3	18.5	15.5	10.8	6.4		
		MCDB	6.9	8.9	14.6	17.7	22.3	26.4	27.6	27.3	23.0	18.1	12.3	7.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	11.3	12.5	11.5	11.1	11.3	11.5	11.2	10.8	9.6	8.9	9.7		
		MCDBR	14.0	15.6	16.9	14.9	13.8	12.9	12.8	12.9	12.7	12.0	11.7	12.4		
	5% WB	MCWBR	9.4	10.1	9.4	7.5	6.3	5.5	5.0	4.7	5.3	6.5	7.7	8.5		
		MCWBR	11.0	12.9	14.3	12.8	12.3	11.7	11.3	11.6	10.7	9.3	8.3	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.339	0.370	0.421	0.467	0.469	0.489	0.469	0.463	0.452	0.434	0.375	0.334		
	taud		2.344	2.252	2.135	2.048	2.101	2.089	2.147	2.168	2.159	2.160	2.288	2.374		
	Ebn at Noon		748	790	796	792	806	789	800	788	757	702	691	713		
	Edh at Noon		78	103	133	158	155	157	147	139	129	112	82	69		
(44) All-Sky Solar Radiation	RadAvg		1.47	2.28	3.60	4.50	5.46	6.31	6.42	5.44	4.09	2.44	1.45	1.22		
	RadStd		0.21	0.32	0.47	0.53	0.45	0.39	0.41	0.35	0.29	0.37	0.25	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.57	N/A	N/A	+0.93	N/A	N/A	N/A	-129	+147	+82
(47) Regional (83 neighbors)		+0.32	N/A	N/A	+0.70	+0.42	+0.33	N/A	N/A	+58	+12

Nomenclature: See separate page